

```
import cv2
import numpy as np
from google.colab import files
from matplotlib import pyplot as plt

# Upload the image
uploaded = files.upload()

# Get the uploaded image name
filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, 0)

# Check if image is loaded
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Apply Sobel Y edge detection
    sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=5)

    # Convert to 8-bit image
    sobel_y = cv2.convertScaleAbs(sobel_y)

    # Save the output
    cv2.imwrite("sobel_y.jpg", sobel_y)

    # Display original image
    plt.figure(figsize=(10, 4))

    plt.subplot(1, 2, 1)
    plt.imshow(img, cmap="gray")
    plt.title("Original Image")
    plt.axis("off")

    # Display Sobel Y output
    plt.subplot(1, 2, 2)
```

```
plt.imshow(sobel_y, cmap="gray")  
plt.title("Sobel Y Edge Detection")  
plt.axis("off")  
  
plt.show()  
  
# Download the output  
files.download("sobel_y.jpg")
```

anime-boy-...ustration.jpg

anime-boy-balcony-watching-sunset-city-illustration.jpg(image/jpeg) - 2747878 bytes, last modified: 8/19/2026 - 100% done
Saving anime-boy-balcony-watching-sunset-city-illustration.jpg to anime-boy-balcony-watching-sunset-city-illust

Original Image



Sobel Y Edge Detection



